

A Generating-Function Bridge from Somos–4 Sequences to Ramanujan Theta Functions

An explicit convolution-free theta expansion

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Abstract

Starting from a generating function whose Hankel determinants satisfy a Somos–4 recurrence, we introduce a multiplicative factor in the argument of a bilateral generating series. This produces a two-variable kernel that admits expansions in Jacobi theta functions and, equivalently, in Ramanujan’s two-variable theta function. The main calculation removes a convolution from the Fourier coefficients and replaces it by an explicit quadratic factor. A classical theta-derivative identity then evaluates the remaining parity scalar as one half of a fourth power of a theta constant. Jacobi’s triple product consequently gives a derivative-free infinite-product factorization of every nonzero antisymmetric Fourier coefficient. Extracting a fixed power of the auxiliary variable then collapses the double theta kernel to a single bilateral theta sum, so the Somos–4 relation becomes an explicit identity among three one-dimensional theta product-sums. The resulting formula gives a direct and computationally usable link among Somos–4 recurrences, generating functions, Hankel determinants, elliptic sigma functions, and Ramanujan theta functions.

1 Introduction and motivation

A Somos–4 sequence is a sequence (H_n) satisfying a fourth-order bilinear recurrence

$$H_{n+4}H_n = \alpha H_{n+3}H_{n+1} + \beta H_{n+2}^2. \quad (1.1)$$

Despite its elementary appearance, this recurrence is closely connected with elliptic curves and elliptic functions. A second line of research begins with an ordinary generating function $G(x) = \sum_{k \geq 0} g_k x^k$ and asks when its Hankel determinants form a Somos–4 sequence. Continued fractions, moment sequences, orthogonal polynomials, and Riordan arrays provide several approaches to that question; see, for example, [1, 2, 9, 10].

This paper develops a complementary generating-function viewpoint. Rather than stopping at the Hankel transform of G , we place the Somos–4 orbit in a bilateral generating series $S(t, q)$ and compare the two arguments tz and t/z . The auxiliary factor z records the difference of two indices: in the product

$$S(tz, q)S(t/z, q), \quad (1.2)$$

the exponent of t records their sum, while the exponent of z records their difference. This simple multiplicative change of argument is the bridge used throughout the paper. After the elliptic sigma representation is converted to theta notation and the nome is aligned, the bridge leads to parity-restricted theta series and then to Ramanujan’s two-variable theta function.

The main result is an explicit expansion whose coefficients contain no unevaluated convolution. It shows how the Somos–4 recurrence is encoded by selected coefficients of a single kernel, and it separates the double series into elementary Fourier coefficients, a quadratic factor, and one parity-dependent scalar.

Terminology and reading guide

We use a few standard terms that may be unfamiliar outside the subject.

- An *ordinary generating function* packages a sequence $(g_k)_{k \geq 0}$ as the formal power series $G(x) = \sum_{k \geq 0} g_k x^k$. Algebraic operations on G can therefore encode identities for its coefficients.
- The *Hankel determinant* $H_n[G]$ is the determinant of the $n \times n$ matrix whose (i, j) entry is g_{i+j} . The sequence $(H_n[G])$ is called the Hankel transform of (g_k) .
- A *kernel* here is simply the two-variable product $\mathcal{K}(t, z; q) = S(tz, q)S(t/z, q)$. The word does not introduce an integral operator; it names the central generating expression from which the Somos–4 relation is extracted.
- A *nome* is the parameter $p = e^{\pi i \tau}$ used in Fourier series for theta functions. “Nome alignment” means choosing the scaling parameter so that the Gaussian factor from the sigma function matches this theta nome.
- A *bilateral series* is summed over all integers rather than only nonnegative integers. A *parity sector* restricts its index to even or odd integers.
- “Convolution-free” means that a coefficient originally written as a sum over products of Fourier coefficients is replaced by a direct formula involving one coefficient and its nome derivative.

Section 1 introduces the generating-function bridge. Sections 2–5 supply the elliptic and Fourier data and derive the explicit coefficient. Sections 6–7 give the double-theta and parity-determinant forms. Section 8 extracts a fixed kernel coefficient and reduces the resulting double sum to one bilateral theta sum. The final sections collect the parameters and state the analytic assumptions.

Generating functions and the surrounding literature

For this paper, generating functions are used formally: $[x^k]G(x) = g_k$ denotes coefficient extraction, and identities are valid coefficient by coefficient whenever the displayed algebraic operations are defined. When numerical or analytic evaluation is intended, convergence assumptions are added explicitly. Useful general introductions are Wilf [14] and Stanley [12]; continued-fraction and Hankel methods are treated by Wall [13], Flajolet [5], and Krattenthaler [7].

In the Somos–4 setting, Barry connected generating functions, continued fractions, Riordan arrays, Hankel transforms, and elliptic curves [1, 2]. Recent work gives sufficient conditions and proofs for broad families of Somos–4 Hankel transforms [9], while [10] addresses the inverse problem of constructing an algebraic generating function from prescribed Somos–4 data. The elliptic sigma-function description of Somos–4 sequences is developed in the work of Hone [6]. For theta functions we use standard Jacobi notation, while Ramanujan’s symmetric function

$$f(a, b) = \sum_{n \in \mathbb{Z}} a^{n(n+1)/2} b^{n(n-1)/2}, \quad |ab| < 1, \quad (1.3)$$

provides the final form of the expansion; classical background is available in Berndt [3] and a concise introduction in Somos [11]. We do not assume a previously established direct Somos–4–Ramanujan identity: the connection claimed here is the explicit bridge derived below through the scaled arguments of the generating series.

2 From the scaled generating function to the kernel

Let

$$G(x) = \sum_{k \geq 0} g_k x^k. \quad (2.1)$$

Its n th Hankel determinant is

$$H_n = H_n[G] := \det(g_{i+j})_{0 \leq i, j \leq n-1}, \quad H_0 := 1. \quad (2.2)$$

Thus the first values are $H_1 = g_0$ and $H_2 = g_0 g_2 - g_1^2$. This example shows that a Hankel determinant combines several coefficients of G into one nonlinear invariant. Under $G(x) \mapsto G(qx)$,

$$H_n[G(qx)] = q^{n(n-1)} H_n[G]. \quad (2.3)$$

Define the scaled orbit and its bilateral generating series by

$$W_n := H_n q^{n(n-1)}, \quad S(t, q) := \sum_{n \in \mathbb{Z}} W_n t^n. \quad (2.4)$$

The word “orbit” refers to the sequence of values produced by iterating the Somos–4 recurrence. Negative indices are included whenever the nondegenerate recurrence determines them. Now set

$$\boxed{\mathcal{K}(t, z; q_*) := S(tz, q_*) S(t/z, q_*)}. \quad (2.5)$$

Equivalently,

$$\mathcal{K}(t, z; q_*) = \sum_{i, j \in \mathbb{Z}} W_i W_j t^{i+j} z^{i-j}. \quad (2.6)$$

If

$$H_{n+4} H_n = \alpha H_{n+3} H_{n+1} + \beta H_{n+2}^2, \quad (2.7)$$

then

$$[z^4] \mathcal{K} = \alpha q_*^6 [z^2] \mathcal{K} + \beta q_*^8 [z^0] \mathcal{K}. \quad (2.8)$$

3 Elliptic data and Fourier nome alignment

The Weierstrass sigma function is an entire function associated with an elliptic lattice; it provides a convenient closed form for generic Somos–4 orbits. We assume the standard nondegenerate sigma representation

$$\boxed{H_n = A_S B_S^n \frac{\sigma(z_0 + n\kappa)}{\sigma(\kappa)^{n^2}}}. \quad (3.1)$$

Here A_S and B_S are nonzero gauge factors, z_0 is the initial elliptic phase, and κ is the fixed translation step. “Nondegenerate” means that the denominators displayed below do not vanish.

Choose fundamental half-periods ω_1, ω_3 and put

$$\tau = \frac{\omega_3}{\omega_1}, \quad p = e^{\pi i \tau}, \quad \lambda = \frac{\pi}{2\omega_1}, \quad a = \frac{\eta_1}{2\omega_1}, \quad (3.2)$$

where $\eta_1 = \zeta(\omega_1)$. We use

$$\boxed{\sigma(u) = C_\sigma e^{au^2} \theta_1(\lambda u, p)}, \quad \boxed{C_\sigma = \frac{2\omega_1}{\pi \theta_1'(0, p)}}. \quad (3.3)$$

Define

$$Q := \frac{q}{\sigma(\kappa)}, \quad \gamma := \frac{\lambda \kappa}{2}, \quad x_0 := \lambda z_0, \quad (3.4)$$

and

$$P := Qe^{a\kappa^2}. \quad (3.5)$$

For the infinite Fourier expansion we impose the alignment

$$\boxed{P^2 = p.} \quad (3.6)$$

After choosing a branch of $p^{1/2}$, set

$$\boxed{q_* := \sigma(\kappa)p^{1/2}e^{-a\kappa^2}}, \quad \boxed{Q_* = p^{1/2}e^{-a\kappa^2}, \quad P = p^{1/2}}. \quad (3.7)$$

Finally put

$$\boxed{X_* := \frac{B_{St}}{q_*}e^{2az_0\kappa}.} \quad (3.8)$$

This is the Fourier alignment used below; in general it differs from alignments imposed after a finite torsion dissection.

4 Parity theta functions

A theta function is a rapidly convergent bilateral series with a quadratic exponent. We separate such a series into its even and odd index parts because the Fourier coefficients below have different closed forms in the two parities.

For $\varepsilon \in \{0, 1\}$ define

$$\boxed{\Theta_\varepsilon(Y; p^{1/2}) := \sum_{\substack{n \in \mathbb{Z} \\ n \equiv \varepsilon \pmod{2}}} p^{n^2/4} Y^n.} \quad (4.1)$$

With

$$\Phi_\rho(\xi) := \sum_{k \in \mathbb{Z}} \rho^{k^2} \xi^k = f(\rho\xi, \rho/\xi), \quad (4.2)$$

one has

$$\boxed{\Theta_\varepsilon(Y; p^{1/2}) = p^{\varepsilon^2/4} Y^\varepsilon \Phi_p(p^\varepsilon Y^2).} \quad (4.3)$$

5 Fourier coefficients and elimination of the convolution

A Fourier coefficient is the multiplier of one exponential mode $e^{2i\ell x}$. Multiplying two Fourier series normally produces a convolution, namely a sum over all pairs of indices with a fixed total. The point of this section is to replace that sum by a direct differential formula.

Define a_ℓ and b_ℓ by

$$\theta_1(x, p)^2 = \sum_{\ell \in \mathbb{Z}} a_\ell(p) e^{2i\ell x}, \quad (5.1)$$

$$\lambda^2(\theta'_1(x, p)^2 - \theta_1(x, p)\theta''_1(x, p)) - 2a\theta_1(x, p)^2 = \sum_{\ell \in \mathbb{Z}} b_\ell(p) e^{2i\ell x}. \quad (5.2)$$

The first sequence has the parity formulas

$$\boxed{a_{2j} = p^{2j^2} \theta_2(0, p^2)}, \quad \boxed{a_{2j+1} = -p^{(2j+1)^2/2} \theta_3(0, p^2)}. \quad (5.3)$$

Equivalently,

$$a_\ell = (-1)^\ell \sum_{n \in \mathbb{Z}} p^{(n+1/2)^2 + (\ell-n-1/2)^2}. \quad (5.4)$$

Let

$$\boxed{D_p := p \frac{d}{dp}}. \quad (5.5)$$

Proposition 5.1 (Convolution-free formula for b_ℓ). *For every $\ell \in \mathbb{Z}$,*

$$\boxed{b_\ell = (4\lambda^2 D_p - 2\lambda^2 \ell^2 - 2a) a_\ell.} \quad (5.6)$$

Proof. The direct convolution is

$$b_\ell = 2\lambda^2 (-1)^\ell \sum_{n \in \mathbb{Z}} (2n+1-\ell)^2 p^{(n+1/2)^2 + (\ell-n-1/2)^2} - 2a a_\ell. \quad (5.7)$$

Set $r = 2n+1-\ell$. Then

$$2(2n+1-\ell)^2 = 2r^2, \quad (5.8)$$

$$(n+\tfrac{1}{2})^2 + (\ell-n-\tfrac{1}{2})^2 = \frac{\ell^2 + r^2}{2}, \quad (5.9)$$

and $r \equiv 1-\ell \pmod{2}$. Hence

$$a_\ell = (-1)^\ell \sum_{\substack{r \in \mathbb{Z} \\ r \equiv 1-\ell \pmod{2}}} p^{(\ell^2 + r^2)/2}. \quad (5.10)$$

Applying D_p and eliminating the r^2 -weighted sum gives

$$(-1)^\ell \sum_r 2r^2 p^{(\ell^2 + r^2)/2} = (4D_p - 2\ell^2) a_\ell, \quad (5.11)$$

which proves the claim. $\square$

Let $\pi(\ell) \in \{0, 1\}$ denote the numerical parity of ℓ , so that $\pi(\ell) = 0$ for even ℓ and $\pi(\ell) = 1$ for odd ℓ , and put

$$T_0(p) := \theta_2(0, p^2), \quad T_1(p) := \theta_3(0, p^2). \quad (5.12)$$

Then

$$\boxed{b_\ell = a_\ell [4\lambda^2 D_p \log T_{\pi(\ell)}(p) - 2a]}. \quad (5.13)$$

Thus b_ℓ contains no convolution and no data beyond the elliptic lattice.

6 Antisymmetric coefficient and its cancellations

Antisymmetrization means replacing a two-index expression $c_{\ell, m}$ by $c_{\ell, m} - c_{m, \ell}$. This forces vanishing on the diagonal $m = \ell$ and, in the present calculation, also removes the Gaussian lattice constant.

Define the single parity modulus

$$\boxed{\Delta(p) := D_p \log \frac{\theta_2(0, p^2)}{\theta_3(0, p^2)}}. \quad (6.1)$$

Proposition 6.1 (Explicit antisymmetric coefficient). *For all $\ell, m \in \mathbb{Z}$,*

$$\boxed{a_\ell b_m - b_\ell a_m = 4\lambda^2 a_\ell a_m \Delta(p) (\pi(\ell) - \pi(m))}. \quad (6.2)$$

Proof. Insert the preceding formula for b_k . The terms $-2a a_k$ cancel identically. The difference of the two parity logarithmic derivatives is $-\Delta$, 0, or Δ according to the parities of (ℓ, m) , which gives the displayed formula. $\square$

Proposition 6.2 (Evaluation of the parity modulus). *For $|p| < 1$ one has*

$$\Delta(p) = \frac{1}{2}\theta_4(0, p^2)^4. \quad (6.3)$$

Consequently,

$$a_\ell b_m - b_\ell a_m = 2\lambda^2 \theta_4(0, p^2)^4 a_\ell a_m (\pi(\ell) - \pi(m)). \quad (6.4)$$

Proof. For the nome convention used here, the classical identity

$$q \frac{d}{dq} \log \frac{\theta_2(0, q)}{\theta_3(0, q)} = \frac{1}{4} \theta_4(0, q)^4 \quad (6.5)$$

holds; see [4, Sec. 20.7(vii)]. Put $q = p^2$. Since $p \, d/dp = 2q \, d/dq$, the definition of $\Delta(p)$ gives

$$\Delta(p) = 2q \frac{d}{dq} \log \frac{\theta_2(0, q)}{\theta_3(0, q)} = \frac{1}{2} \theta_4(0, q)^4. \quad (6.6)$$

Replacing q by p^2 proves the first formula, and substitution in the preceding proposition proves the second. $\square$

As a normalization check, when $p \rightarrow 0$ one has $\theta_2(0, p^2)/\theta_3(0, p^2) \sim 2p^{1/2}$, so both sides of the first formula tend to $1/2$.

Let

$$(x; q)_\infty := \prod_{n=0}^{\infty} (1 - xq^n). \quad (6.7)$$

Jacobi's triple product gives

$$\begin{aligned} \theta_2(0, p^2) &= 2p^{1/2} (p^4; p^4)_\infty (-p^4; p^4)_\infty^2, \\ \theta_3(0, p^2) &= (p^4; p^4)_\infty (-p^2; p^4)_\infty^2, \\ \theta_4(0, p^2) &= (p^4; p^4)_\infty (p^2; p^4)_\infty^2. \end{aligned} \quad (6.8)$$

Hence

$$\Delta(p) = \frac{1}{2} (p^4; p^4)_\infty^4 (p^2; p^4)_\infty^8. \quad (6.9)$$

6.1 Arithmetic coefficient sequence of the parity modulus

The fourth power in the evaluation of $\Delta(p)$ also gives an explicit arithmetic interpretation of its power-series coefficients. Define

$$A(n) := \sum_{\substack{d|n \\ 4 \nmid d}} d = \sigma_1(n) - 4\sigma_1(n/4), \quad (6.10)$$

where the second term is understood to be zero unless $4 \mid n$. The first values are

$$1, 3, 4, 3, 6, 12, 8, 3, 13, 18, 12, 12, \dots, \quad (6.11)$$

and this sequence is OEIS A046897 [8].

The ordinary generating function recorded there is

$$\frac{\theta_3(0, x)^4 - 1}{8} = \sum_{n \geq 1} A(n) x^n. \quad (6.12)$$

This is the classical four-square identity in coefficient form. Since $\theta_4(0, q) = \theta_3(0, -q)$, substitution of $x = -p^2$ into this identity, together with the already proved formula $\Delta(p) = \frac{1}{2}\theta_4(0, p^2)^4$, yields

$$\Delta(p) = \frac{1}{2} + 4 \sum_{n \geq 1} (-1)^n A(n) p^{2n}. \quad (6.13)$$

Thus

$$\Delta(p) = \frac{1}{2} - 4p^2 + 12p^4 - 16p^6 + 12p^8 - 24p^{10} + 48p^{12} - 32p^{14} + \dots \quad (6.14)$$

Equivalently, for $n \geq 1$,

$$[p^{2n}] \Delta(p) = 4(-1)^n \sum_{\substack{d|n \\ 4 \nmid d}} d, \quad [p^{2n+1}] \Delta(p) = 0. \quad (6.15)$$

Thus every nonzero opposite-parity coefficient is an elementary power of p times one universal infinite product. Explicitly,

$$a_{2r} b_{2s+1} - b_{2r} a_{2s+1} = 4\lambda^2 p^{2r^2 + (2s+1)^2/2 + 1/2} \times (p^4; p^4)_\infty^6 (p^2; p^4)_\infty^8 (-p^4; p^4)_\infty^2 (-p^2; p^4)_\infty^2. \quad (6.16)$$

Antisymmetry gives the odd-even sector, and the same-parity sectors vanish. After the elementary power of p is removed, the remaining product is independent of r and s .

The four parity sectors are therefore

$(\pi(\ell), \pi(m))$	$(a_\ell b_m - b_\ell a_m) / (4\lambda^2 a_\ell a_m)$
$(0, 0)$	0
$(1, 1)$	0
$(0, 1)$	$\Delta(p)$
$(1, 0)$	$-\Delta(p)$

(6.17)

In particular, every same-parity term contains

$$a_\ell b_m - b_\ell a_m = 0 \quad (\pi(\ell) = \pi(m)), \quad (6.18)$$

so the same-parity contribution vanishes on both the diagonal $m = \ell$ and the anti-diagonal $m = -\ell$. The full coefficient always vanishes on $m = \ell$ by antisymmetry.

7 Convolution-free infinite-theta kernel

A rank-two combination here means a difference of two separable products, one depending on ℓ and the other on m . The kernel is obtained from this antisymmetric combination by the normalization

$$S(t, q_*) = A_S C_\sigma e^{az_0^2} \sum_{r \in \mathbb{Z}} c_r e^{i(2r+1)x_0} \Phi_{p^{1/2}} \left(X_* e^{2i(2r+1)\gamma} \right), \quad c_r := -i(-1)^r p^{(r+1/2)^2}. \quad (7.1)$$

The factor $-1/2$ follows from the antisymmetrization convention: the ordered rank-two difference counts the two exchanged terms separately, while their order is opposite to the orientation in the product $S(tz, q_*) S(t/z, q_*)$. Substitution of the explicit antisymmetric coefficient gives

$$\begin{aligned} \mathcal{K}(t, z; q_*) &= A_S^2 C_\sigma^2 e^{2az_0^2} \sum_{r, s \in \mathbb{Z}} c_r c_s e^{2i(r+s+1)x_0} \\ &\quad \times \Phi_{p^{1/2}} \left(X_* z e^{2i(2r+1)\gamma} \right) \Phi_{p^{1/2}} \left(X_* z^{-1} e^{2i(2s+1)\gamma} \right). \end{aligned} \quad (7.2)$$

Expanding the two parity theta functions gives the corresponding Ramanujan form

$$\begin{aligned} \mathcal{K}(t, z; q_*) &= A_S^2 C_\sigma^2 e^{2az_0^2} \sum_{r,s \in \mathbb{Z}} c_r c_s e^{2i(r+s+1)x_0} \\ &\quad \times f\left(p^{1/2} X_* z e^{2i(2r+1)\gamma}, \frac{p^{1/2}}{X_* z e^{2i(2r+1)\gamma}}\right) \\ &\quad \times f\left(p^{1/2} X_* z^{-1} e^{2i(2s+1)\gamma}, \frac{p^{1/2}}{X_* z^{-1} e^{2i(2s+1)\gamma}}\right). \end{aligned} \quad (7.3)$$

8 Single-sum formulas for the kernel coefficients

The preceding double-theta expansion describes the complete dependence on the auxiliary variable z . For the Somos–4 relation, however, only the three coefficients of z^0 , z^2 , and z^4 are needed. Extracting a fixed coefficient before performing the outer sums produces a further reduction. The argument uses only the already defined function Φ_ρ and introduces one parity selector for a Jacobi theta factor.

Proposition 8.1 (Coefficient extraction for a pair of bilateral theta series). *For nonzero A and B and every $k \in \mathbb{Z}$,*

$$[z^k] \left(\Phi_{p^{1/2}}(Az) \Phi_{p^{1/2}}(Bz^{-1}) \right) = A^k p^{k^2/2} \Phi_p(p^k AB). \quad (8.1)$$

Proof. Expanding the two bilateral series gives

$$\Phi_{p^{1/2}}(Az) \Phi_{p^{1/2}}(Bz^{-1}) = \sum_{u,v \in \mathbb{Z}} p^{(u^2+v^2)/2} A^u B^v z^{u-v}. \quad (8.2)$$

The coefficient of z^k is obtained by setting $u = v + k$. Hence

$$\begin{aligned} [z^k] \left(\Phi_{p^{1/2}}(Az) \Phi_{p^{1/2}}(Bz^{-1}) \right) &= \sum_{v \in \mathbb{Z}} p^{((v+k)^2+v^2)/2} A^{v+k} B^v \\ &= A^k p^{k^2/2} \sum_{v \in \mathbb{Z}} p^{v^2} (p^k AB)^v, \end{aligned} \quad (8.3)$$

which is the stated formula. □

To apply this identity to the kernel, define only the following parity-selected factor:

$$\mathcal{J}_k(h; p) := \begin{cases} \theta_2(2k\gamma, p^2), & h \equiv 0 \pmod{2}, \\ \theta_3(2k\gamma, p^2), & h \equiv 1 \pmod{2}. \end{cases} \quad (8.4)$$

This notation is separate from the theta constants $T_0(p)$ and $T_1(p)$ used in Section 4: the latter occur only at zero argument, whereas $\mathcal{J}_k(h; p)$ records both the extracted coefficient k and the parity of the remaining summation index h .

Proposition 8.2 (Single-sum coefficient formula). *For every $k \in \mathbb{Z}$,*

$$\begin{aligned} [z^k] \mathcal{K}(t, z; q_*) &= A_S^2 C_\sigma^2 e^{2az_0^2} X_*^k p^{k^2/2} \sum_{h \in \mathbb{Z}} (-1)^h p^{h^2/2} e^{2ih(x_0+k\gamma)} \\ &\quad \times \mathcal{J}_k(h; p) \Phi_p(p^k X_*^2 e^{4ih\gamma}). \end{aligned} \quad (8.5)$$

Proof. In the double-theta kernel of Section 6, apply the preceding proposition with

$$A = X_* e^{2i(2r+1)\gamma}, \quad B = X_* e^{2i(2s+1)\gamma}. \quad (8.6)$$

This gives

$$\begin{aligned} [z^k]\mathcal{K} &= A_S^2 C_\sigma^2 e^{2az_0^2} X_*^k p^{k^2/2} \sum_{r,s \in \mathbb{Z}} c_r c_s e^{2i(r+s+1)x_0} e^{2ik(2r+1)\gamma} \\ &\quad \times \Phi_p\left(p^k X_*^2 e^{4i(r+s+1)\gamma}\right). \end{aligned} \quad (8.7)$$

Set $h = r + s + 1$, so that $s = h - 1 - r$. The last theta factor now depends only on h . From the definition of c_r ,

$$c_r c_{h-1-r} = (-1)^h p^{(r+1/2)^2 + (h-r-1/2)^2}. \quad (8.8)$$

With

$$y = r + \frac{1-h}{2}, \quad (8.9)$$

one has

$$(r + \frac{1}{2})^2 + (h - r - \frac{1}{2})^2 = \frac{h^2}{2} + 2y^2, \quad 2r + 1 = h + 2y. \quad (8.10)$$

If h is even, then $y \in \mathbb{Z} + 1/2$; if h is odd, then $y \in \mathbb{Z}$. Therefore

$$\sum_{r \in \mathbb{Z}} c_r c_{h-1-r} e^{2ik(2r+1)\gamma} = (-1)^h p^{h^2/2} e^{2ikh\gamma} \mathcal{J}_k(h; p). \quad (8.11)$$

Substitution gives the claimed single-sum formula. $\square$

The remaining factor has an equivalent Ramanujan form because $\Phi_p(\xi) = f(p\xi, p/\xi)$:

$$\Phi_p\left(p^k X_*^2 e^{4ih\gamma}\right) = f\left(p^{k+1} X_*^2 e^{4ih\gamma}, p^{1-k} X_*^{-2} e^{-4ih\gamma}\right). \quad (8.12)$$

The product of the two arguments is always p^2 . Thus the coefficient reduction changes the elliptic argument but not the nome base of Ramanujan's theta function.

Jacobi's triple product also gives the product form

$$\boxed{\begin{aligned} \Phi_p\left(p^k X_*^2 e^{4ih\gamma}\right) &= (p^2; p^2)_\infty \left(-p^{k+1} X_*^2 e^{4ih\gamma}; p^2\right)_\infty \\ &\quad \times \left(-p^{1-k} X_*^{-2} e^{-4ih\gamma}; p^2\right)_\infty. \end{aligned}} \quad (8.13)$$

Consequently, each coefficient $[z^k]\mathcal{K}$ is a one-dimensional sum of explicit theta products rather than the double sum of Section 6.

For a concise statement of the induced Somos-4 identity, define

$$\boxed{\begin{aligned} \mathcal{L}_k(t) &:= X_*^k p^{k^2/2} \sum_{h \in \mathbb{Z}} (-1)^h p^{h^2/2} e^{2ih(x_0 + k\gamma)} \mathcal{J}_k(h; p) \\ &\quad \times \Phi_p\left(p^k X_*^2 e^{4ih\gamma}\right). \end{aligned}} \quad (8.14)$$

The variable t enters through X_* , as defined in Section 3. The single-sum formula is equivalently

$$[z^k]\mathcal{K}(t, z; q_*) = A_S^2 C_\sigma^2 e^{2az_0^2} \mathcal{L}_k(t). \quad (8.15)$$

The common prefactor cancels from the coefficient recurrence, giving the following direct theta identity.

Corollary 8.3 (Single-sum Somos-4 identity). *The three one-dimensional theta sums satisfy*

$$\boxed{\mathcal{L}_4(t) = \alpha q_*^6 \mathcal{L}_2(t) + \beta q_*^8 \mathcal{L}_0(t).} \quad (8.16)$$

Proof. Substitute the preceding expression for $[z^k]\mathcal{K}$ with $k = 0, 2, 4$ into the kernel coefficient recurrence from Section 2 and cancel $A_S^2 C_\sigma^2 e^{2az_0^2}$. $\square$

The reduction is also suited to finite dissections. If $e^{4iN\gamma} = 1$ for a positive integer N , the factor $e^{4ih\gamma}$ depends only on the residue class of h modulo N . Writing $h = Nj + r$ therefore separates the single sum into N residue-class sums. This observation supplies a direct starting point for torsion specializations, but no additional torsion assumption is required for the identities proved above.

9 A two-component parity-determinant reduction

Proposition 9.1 (Rank-two parity expansion). *With the notation of Sections 3–5,*

$$\boxed{\mathcal{K}(t, z; q_*) = A_S^2 C_\sigma^4 e^{2az_0^2} \sum_{\varepsilon=0}^1 \sum_{\ell, m \in \mathbb{Z}} e^{2i\ell x_0} (a_\ell b_m - b_\ell a_m) \times \Theta_\varepsilon(X_* e^{2i\ell\gamma}; p^{1/2}) \Theta_\varepsilon(z e^{2im\gamma}; p^{1/2}).} \quad (9.1)$$

Proof. Start from

$$\mathcal{K}(t, z; q_*) = \sum_{i, j \in \mathbb{Z}} W_i W_j t^{i+j} z^{i-j} \quad (9.2)$$

and introduce

$$u = i + j, \quad v = i - j, \quad i = \frac{u+v}{2}, \quad j = \frac{u-v}{2}. \quad (9.3)$$

Thus u and v have the same parity. The (u, v) -lattice is therefore the disjoint union of $u \equiv v \equiv \varepsilon \pmod{2}$, with $\varepsilon \in \{0, 1\}$. Insert the aligned sigma representation for W_i and W_j , apply the rank-two theta addition identity, and expand its two theta factors in the Fourier bases defining a_ℓ and b_ℓ . The Fourier pair (ℓ, m) then carries the alternating coefficient

$$\det \begin{pmatrix} a_\ell & a_m \\ b_\ell & b_m \end{pmatrix} = a_\ell b_m - b_\ell a_m. \quad (9.4)$$

For fixed ε , the sums over u and v are precisely the two parity theta functions displayed in the proposition. The remaining phase is $e^{2i\ell x_0}$, and the sigma normalization supplies $A_S^2 C_\sigma^4 e^{2az_0^2}$.

This argument is coefficientwise: after the parity theta functions are expanded, the coefficient of $X_*^u z^v$ vanishes unless $u \equiv v \pmod{2}$. In the surviving sector the inverse change of variables gives the unique integers $i = (u+v)/2$ and $j = (u-v)/2$, and the addition identity reconstructs $W_i W_j$. This is also the finite-truncation check used in the companion computation. $\square$

Substituting Proposition 5.1 gives

$$\boxed{\mathcal{K}(t, z; q_*) = 4\lambda^2 \Delta(p) A_S^2 C_\sigma^4 e^{2az_0^2} \sum_{\varepsilon=0}^1 \sum_{\ell, m \in \mathbb{Z}} e^{2i\ell x_0} a_\ell a_m (\pi(\ell) - \pi(m)) \times \Theta_\varepsilon(X_* e^{2i\ell\gamma}; p^{1/2}) \Theta_\varepsilon(z e^{2im\gamma}; p^{1/2}).} \quad (9.5)$$

This separates naturally into even and odd Fourier sectors.

For $\varepsilon, \nu \in \{0, 1\}$ define

$$\boxed{\mathcal{R}_{\varepsilon, \nu}(t) := \sum_{\substack{\ell \in \mathbb{Z} \\ \pi(\ell) = \nu}} e^{2i\ell x_0} a_\ell \Theta_\varepsilon \left(X_* e^{2i\ell \gamma}; p^{1/2} \right),} \quad (9.6)$$

$$\boxed{\mathcal{D}_{\varepsilon, \nu}(z) := \sum_{\substack{m \in \mathbb{Z} \\ \pi(m) = \nu}} a_m \Theta_\varepsilon \left(z e^{2im\gamma}; p^{1/2} \right).} \quad (9.7)$$

Proposition 9.2 (Parity-determinant reduction). *With the preceding notation,*

$$\boxed{\mathcal{K}(t, z; q_*) = 2\lambda^2 \theta_4(0, p^2)^4 A_S^2 C_\sigma^4 e^{2az_0^2} \sum_{\varepsilon=0}^1 \left(\mathcal{R}_{\varepsilon, 1}(t) \mathcal{D}_{\varepsilon, 0}(z) - \mathcal{R}_{\varepsilon, 0}(t) \mathcal{D}_{\varepsilon, 1}(z) \right).} \quad (9.8)$$

Proof. Insert the explicit antisymmetric coefficient into the parity-theta double series above. The factor $\pi(\ell) - \pi(m)$ vanishes unless ℓ and m have opposite parity. In the even–odd sector it contributes -1 , and in the odd–even sector it contributes $+1$. Grouping those two sectors yields the displayed determinant. $\square$

The kernel coefficients still satisfy the Somos–4 relation. Extracting z^k from $\sum_{i, j \in \mathbb{Z}} W_i W_j t^{i+j} z^{i-j}$ imposes $i - j = k$. Writing $j = n$ and $i = n + k$ gives

$$\boxed{[z^k] \mathcal{K}(t, z; q_*) = \sum_{n \in \mathbb{Z}} W_{n+k} W_n t^{2n+k}.} \quad (9.9)$$

Corollary 9.3 (Coefficient recurrence). *One has*

$$\boxed{[z^4] \mathcal{K} = \alpha q_*^6 [z^2] \mathcal{K} + \beta q_*^8 [z^0] \mathcal{K}.} \quad (9.10)$$

Proof. This is the coefficientwise image of the Somos–4 recurrence

$$W_{n+4} W_n = \alpha q_*^6 W_{n+3} W_{n+1} + \beta q_*^8 W_{n+2}^2. \quad (9.11)$$

$\square$

10 Free and derived quantities

Symbol	Status	Meaning / determination
$G(x)$	input	Original moment generating function. Its Hankel determinants give H_n .
q_*	Fourier alignment	$q_* = \sigma(\kappa) p^{1/2} e^{-a\kappa^2}$ after choosing a branch of $p^{1/2}$.
A_S, B_S, z_0, κ	orbit data	Gauge constants, initial elliptic phase, and elliptic translation step.
ω_1, ω_3	lattice data	Fundamental half-periods; $\tau = \omega_3/\omega_1$.
p, λ, a, C_σ	derived lattice data	Defined by the sigma/theta normalization above.

P	aligned value	$P = p^{1/2}$, so the outer bilateral theta functions have nome $P^2 = p$.
γ, x_0, X_*	derived	Shift phase, initial phase, and the combined t -variable.
a_ℓ	explicit coefficient	Fourier coefficient of θ_1^2 , with closed parity formulas.
b_ℓ	eliminated coefficient	$b_\ell = (4\lambda^2 D_p - 2\lambda^2 \ell^2 - 2a)a_\ell$; no convolution is needed.
$\mathcal{J}_k(h; p)$	derived parity factor	Selects $\theta_2(2k\gamma, p^2)$ for even h and $\theta_3(2k\gamma, p^2)$ for odd h in the single-sum coefficient formula.
$\mathcal{L}_k(t)$	derived coefficient sum	One-dimensional theta sum with $[z^k]\mathcal{K} = A_S^2 C_\sigma^2 e^{2az_0^2} \mathcal{L}_k(t)$; it satisfies the induced Somos-4 identity for $k = 0, 2, 4$.
$\Delta(p)$	derived lattice scalar	$\Delta(p) = \frac{1}{2}\theta_4(0, p^2)^4 = \frac{1}{2}(p^4; p^4)_\infty^4 (p^2; p^4)_\infty^8$; the parity coupling is derivative-free and has an explicit product form.

11 Analytic interpretation

The identities may be read coefficientwise in a formal setting. For an analytic interpretation, assume $|p| < 1$ and restrict the remaining variables to a domain on which the bilateral theta sums and their differentiated series converge absolutely and locally uniformly. The operators D_p , ∂_x^2 , and ∂_y^2 may then be applied term by term.

12 Conclusion

The infinite theta expansion has a convolution-free coefficient structure. The sequence b_ℓ is generated from a_ℓ by a quadratic spectral operator and a nome derivative. Antisymmetrization removes the Gaussian constant, and the remaining parity modulus evaluates to $\Delta(p) = \frac{1}{2}\theta_4(0, p^2)^4$. Its nonconstant coefficients form a signed and rescaled copy of OEIS A046897, equivalently the sums of divisors not divisible by 4, and Jacobi's four-square theorem supplies a lattice-point interpretation. Jacobi's triple product then makes every nonzero opposite-parity coefficient an elementary power of p times one universal infinite product, while the two opposite-parity sectors remain coupled with opposite signs. After the rank-two normalization is included, the kernel is expressed both as an explicit double-theta series and as a two-component parity determinant. Extracting a fixed power of the auxiliary variable collapses the double series to one bilateral theta sum, and the Somos-4 recurrence becomes the explicit identity $\mathcal{L}_4 = \alpha q_*^6 \mathcal{L}_2 + \beta q_*^8 \mathcal{L}_0$. The single-sum form also isolates the residue-class dependence needed for finite torsion dissections. These forms provide direct starting points for modular identities and special-value reductions. In particular, the multiplicative factor in the arguments of the generating series turns a Somos-4 Hankel orbit into a coefficient-selecting kernel, and nome alignment then converts that kernel into Jacobi and Ramanujan theta expansions.

Disclosure on authoring and verification tools

The text of this manuscript was revised and partially rewritten with the assistance of AI-based language tools. The purpose was to improve clarity and place the mathematical material into proper L^AT_EX while preserving the technical content and the mathematical claims. The author remains responsible for the final text, all statements, and the correctness of the formulas.

AI-assisted workflows were also used to draft and test PARI/GP scripts for verifying the identities and formulas appearing in this paper. PARI/GP merits explicit recognition as a powerful and highly effective tool for exact symbolic and arithmetic verification, and it played a useful role in checking the computations underlying the present work.

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